

What is Encryption?

By: [Me]

Without encryption, checking your bank account, sending a message, or making a video call would be exposed to anyone willing to intercept them.

Encryption is what makes keeps the internet safe.



Symmetric Encryption

- Encryption is the act of encrypting messages (plaintext) with a key into indecipherable ciphertext and decrypting that ciphertext back into plaintext
- Symmetric Encryption uses a shared key to encrypt messages (plaintext) into ciphertext and decrypt back into plaintext
- Passed physically between two parties
- Required a new encryption scheme with the advent of the internet



Public Key Cryptography



Diffie-Hellman Key Exchange

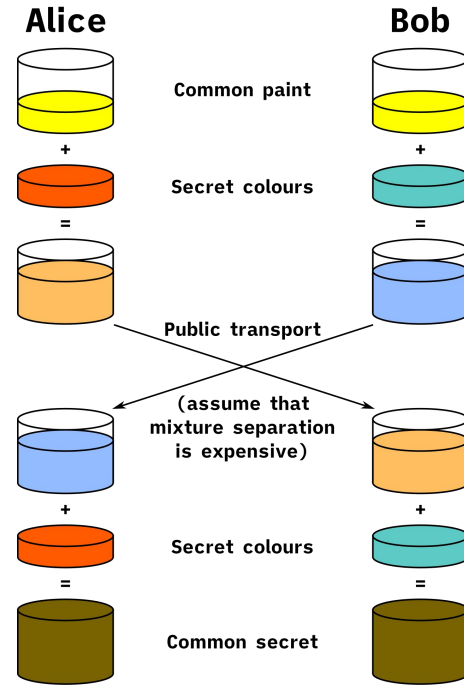


Figure 1: Diffie-Hellman Key Exchange. [Reference](#)

Hybrid Approach 2



Applications

- Web Browsing (HTTPS, TLS)
- Secure communications (WhatsApp, Signal)
- Digital Signatures (Certificates)
- Authentication (Banking Account, Login Screens)
- Blockchain technology (Crypto)



Next week I plan on extending the research into how the TLS protocol works using the encryption hybrid approach



References

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Questions?